



1,3,2-Dioxathiolane 2,2-dioxide (DTD)

ELECTROLYTE ADDITIVE

PRODUCT OVERVIEW

1,3,2-Dioxathiolane 2,2-dioxide (DTD) is an electrolyte additive supplied for battery-electrolyte formulation, additive screening, and electrochemical research. This sheet summarizes specification limits and representative quality-control results.

PRODUCT INFORMATION

CAS No. 1072-53-3 | Formula $C_2H_4O_4S$ | Abbreviation DTD

TECHNICAL SPECIFICATIONS

PROPERTY	SPECIFICATION	REPRESENTATIVE QC RESULT	UNIT
Appearance	White crystalline material or powder, no mechanical impurities	White crystalline material or powder, no mechanical impurities	—
Assay	≥99.5	99.8074	wt%
Water (Karl Fischer titration)	≤100	38.1	µg/g
Acidity, as HF	≤100	25.8	ppm
Color	≤30	22.4	Hazen
Ethylene sulfite	≤500	0	ppm
Ethylene glycol	≤300	0	ppm
Chloride	≤5	0.73	µg/g

REPRESENTATIVE QC HIGHLIGHTS

99.8074 wt% assay

38.1 µg/g water

25.8 ppm acidity

RESEARCH APPLICATIONS

Electrolyte formulation development; additive screening; cell-material compatibility studies; and electrochemical research. Suitability should be confirmed in the intended system.

STORAGE / SHELF LIFE

90 days

QUALITY / DOCUMENTATION

Specification limits and representative quality-control results are provided for technical reference. Lot-specific Certificate of Analysis is available upon request.